PROJECT DELIVERY / OCTOBER 2026

# MarketBudget

A clearer plan for the next market week.

| 2025 BASKET MAE | NOMINAL 80% COVERAGE | EVALUATED BASKETS |
|---|---|---|
| LKR 2,674 | 78.1% | 201 |

## The decision this project supports

A small caterer or canteen must reserve cash before next week's Dambulla vegetable prices are known. The buyer enters kilograms of beans, carrots and tomatoes and an optional spending limit. The application returns a dated reference-cost estimate and an empirical joint range, with explicit source dates and historical-replay status.

## What has been built

A responsive browser dashboard, local FastAPI/SQLite service, saved plans, backup restoration and exports, price explorer, source audit and correction trail, reproducible data pipeline, restricted quantile models, chronological evaluation, release artifacts and tests. Application 1.2 uses transactional IndexedDB in the hosted edition and SQLite locally, with explicit storage selection, strict backup recovery and decimal arithmetic. A separate six-fold research study compares 20 fixed candidates; the approved model above is unchanged.

## The result that governs the release

The strongest development candidate, pooled price/weather LightGBM, improved basket MAE by 16.96%. Its tomato error breached the 5% regression guard, so the release uses the latest-price baseline. No model was chosen using 2025 outcomes. This preserves the honest fallback required by the proposal.

Status: historical research replay through 10 March 2026. Not a current supplier quotation, prospective field trial, invoice-cost model or verified savings claim.

# A traceable foundation

The recovered vegetablesSriLanka 1.1.0 archive contains 125,576 rows. The selected three-series all-year slice contains 6,660 rows and 157 missing quotes. This implementation retains 2,958 rows from 2022 onward, with 2,926 valid prices, through 10 March 2026.

## Corrections without erased history

The supplied proposal did not include its local audit pack or the six disputed 2021 keys. All pre-2022 prices are therefore excluded. Three 11 July 2024 package values (190, 68, 210) were independently checked against the CBSL wholesale table and corrected to 375, 345, 190 LKR/kg. Raw bytes, URLs, hashes and original values are retained.

| Date | Beans | Carrot | Tomato | Evidence |
|---|---|---|---|---|
| 11 Jul 2024 | 375 | 345 | 190 | Primary wholesale Today |
| 10 Mar 2026 | 290 | 225 | 110 | Primary wholesale Today |
| 16 Oct 2026 | 335 | 225 | 375 | Reference only; separate |

Fifteen primary quotes were checked across 2023–2026, including three October reference-only values. The June 2022 sample requests returned 404, so no primary 2022 quote was verified. The audit is partial. Initial parser failures remain in the attempt log; final successful reparses are listed separately.

## Weather is a tested signal

NASA POWER supplies rainfall, mean/minimum/maximum temperature and humidity for Nuwara Eliya, Badulla and Kandy/Matale grid proxies. Each query returned 1,591 days from 1 November 2021 through 10 March 2026. Features end at least seven days before issue. Grid proxies are not farm stations, district averages or verified supply shares. Local Solar Time and archive revisions limit historical availability claims.

## Collection fails closed

The parser validates the printed date, wholesale/retail header order, Dambulla Today alignment, kg units and ten quote cells. Unknown layouts enter a review queue. The collector stages actual linked CBSL reports for review; it does not append, train or publish automatically.

Source rights: package metadata declares GPL-3. Upstream CBSL training and redistribution permission remains unconfirmed; private research access does not resolve those terms.

# Time comes first

## One precise weekly target

For issue date d, average valid wholesale point quotes on d+1 through d+7, excluding issue day. Require at least four observations per vegetable and a full seven-day calendar window. A basket outcome requires all three labels. Missing outcomes are never interpolated. The point basket adds quantity × central price and is not necessarily the basket median.

## Past-only features

Price inputs include latest available price and age, 7/14/28-day last-known lags, observed means and counts, volatility and momentum. Known weekday and annual cycles supply calendar features. Weather summaries are shifted seven days. A future-data mutation test verifies that subsequent values cannot change earlier features.

| Purpose | Period | Labels | Baskets |
|---|---|---|---|
| train | 2022-01-01 to 2023-12-31 | 1205 | 401 |
| tuning | 2024-01-01 to 2024-06-30 | 291 | 97 |
| calibration | 2024-07-01 to 2024-12-31 | 315 | 105 |
| test_2025 | 2025-01-01 to 2025-12-31 | 607 | 201 |
| later_2026 | 2026-01-01 to 2026-03-10 | 123 | 41 |

Targets crossing each period's end date are purged. Candidate methods share eligible comparison rows. Training stays in 2022–2023 after selection; calibration is reserved for intervals. An earlier train-2022/validate-2023 fold and early-2026 check add context without changing final-test selection.

## Seven candidates; one fixed release rule

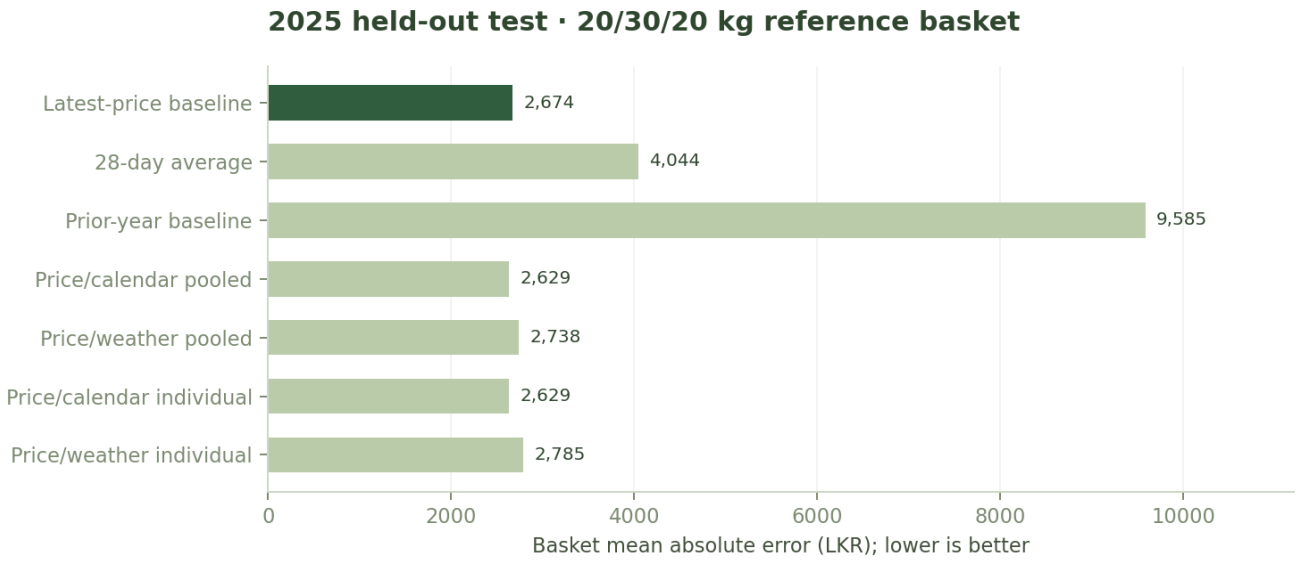
Three baselines (latest price, trailing 28-day mean, prior-year weekly price) compete with pooled/individual LightGBM models using price/calendar features with or without weather. Each learned variant fits quantiles 0.10/0.50/0.90. Relative targets are reconstructed into nonnegative prices; crossing quantiles are sorted.

Fixed LightGBM: 180 trees, learning rate 0.04, 9 leaves, depth 4, minimum leaf samples 30, L2=2, seed 42, two CPU threads. Lowest development basket-MAE learned candidate must improve by ≥10% with no commodity regression above 5%; otherwise use the strongest baseline.

# Evidence, including limits

| Method | 2024 dev MAE | 2025 MAE | Status |
|---|---|---|---|
| Latest-price baseline | 4,560 | 2,674 | Selected |
| 28-day average | 10,037 | 4,044 | Comparison |
| Prior-year baseline | 15,478 | 9,585 | Comparison |
| Price/calendar pooled | 4,033 | 2,629 | Comparison |
| Price/weather pooled | 3,786 | 2,738 | Comparison |
| Price/calendar individual | 4,086 | 2,629 | Comparison |
| Price/weather individual | 3,927 | 2,785 | Comparison |

Selected 2025 test: 201 complete baskets; MAE LKR 2,674.05; signed estimate-minus-outcome bias LKR -52.11. Commodity MAEs: beans 78.60, carrot 62.13, tomato 38.26 LKR/kg. Reference basket: 20 / 30 / 20 kg.




## An informative range can still be wide

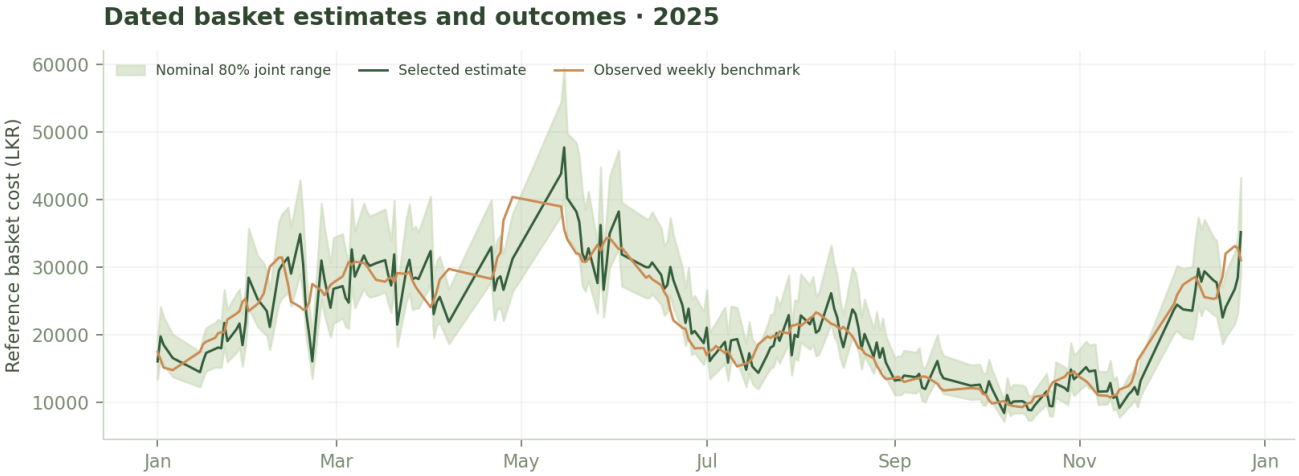
The nominal 80% joint range covered 78.11% of complete 2025 outcomes. Mean width was LKR 8,855.59. The range is derived from 105 aligned July–December 2024 relative residual triplets, preserving commodity co-movement. Conservative order statistics and nonnegative prices are fixed before final testing.

Overlapping labels and calibration targets are not independent weekly trials. The 14-day block-bootstrap selected-minus-baseline error interval is [0,0] because the selected method is that baseline; it does not imply zero forecast error. Learned pinball loss and raw interval coverage are in evaluation.json.

# A complete local workflow

## Evidence to application

Immutable source responses → strict parser and review queue → canonical price/weather tables → as-of features and purged experiments → versioned contracts and joint residual scenarios → browser planner / local FastAPI → saved records and dated exports.




## Usable, inspectable behavior

The dashboard supports editable kilograms, optional budget, price context, cost shares, nominal range, historical date selection, saved baskets, outcome comparisons and source details. Exports retain dates, model version, source hash and replay status. Browser-hosted records stay in browser storage; the local API uses SQLite. Contract arithmetic is shared in definition and checked across Python and JavaScript.

## Freshness is an enforced contract

Current mode requires today's Sri Lankan civil-day issue-date contract and every required price no more than three calendar days old. Stale responses contain no actionable estimate. The current included data therefore stays in replay mode. Missing target outcomes return unavailable, not a successful score.

## Verification

Tests cover exact target windows, missing labels, future-data invariance, weather buffer, boundary purges, malformed/zero baskets, joint residual dependence, freshness boundaries, source table/date parsing, saved-record CRUD, comparison, CSV protection and Python/browser numerical parity. Browser QA exercises the five views at 320, 390, 680, 820, 1024 and 1440 px, plus save/restore/delete and JSON/CSV download.

Run: python -m pip install -e '.[dev]' · python scripts/build.py · python -m marketbudget.api. Open http://127.0.0.1:8000. The API is loopback-only; multi-user authentication and hosted Python execution are outside this release.

# Ready for honest review

## What is complete

A runnable repository, real-data research replay, reproducible preprocessing and experiments, local API and persistence, responsive app, evaluation figures, tests, source attributions, technical documentation, two-page summary and three-minute demonstration script. No source observations, scores, interviews or testimonials were fabricated.

## What is still required for operational use

Confirm upstream price-source terms. Complete source sampling, particularly 2022, and audit a continuous recent extension. Recruit a buyer and verify that a weekly-average target fits their purchasing schedule. Agree acceptable rupee error and range width. Preserve prospective issue snapshots and run a four-to-eight-week shadow pilot before assessing real decision value.

Delivered prices can differ by grade, volume, credit, supplier and transport. No invoice-cost adjustment, demand forecast, automated order or shortage prediction is claimed. Historical publication timestamps and weather vintages are unknown; calibration has only 105 overlapping dates and may fail under regime change.

## Source and software credits

Talagala, Jayarathne and Talagala, vegetablesSriLanka 1.1.0: https://thiyangtdata.r-universe.dev/vegetablesSriLanka (GPL-3 package metadata).

CBSL Daily Price Reports: https://www.cbsl.gov.lk/en/statistics/economic-indicators/price-report. Original source PDFs, correction evidence and SHA-256 manifests retained in the local research project.

NASA POWER daily API: https://power.larc.nasa.gov/docs/services/api/temporal/daily/ and https://power.larc.nasa.gov/docs/faqs/data/ . Regional grid data acknowledged.

LightGBM, pandas, NumPy, FastAPI, Uvicorn, HTTPX, PyMuPDF and pyreadr retain their respective licenses. DM Sans uses the SIL Open Font License. New code is GPL-3.0-or-later; data rights are separate.

## AI assistance and ownership

OpenAI Codex assisted with source recovery, implementation, experiments, tests and documentation. The original team should review and own the submitted work and retain this disclosure. A live buyer pilot and verified savings have not been established.

Detailed numerical evidence: reports/evaluation.json; artifacts/model_card.json; data/processed/source_audit.json; reports/basket_predictions.csv. Reproduction and operations: README.md and docs/OPERATIONS.md.